

Lesson 2.2 – The Seven Pillars

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The complete, defensible answer to “Why do we still need human developers?” rests on seven pillars. Each one is a thing AI cannot credibly do in the next five years. Memorize the pillars. Memorize the soundbites. The body of every AI-question answer in your career builds on these.

By the end of this lesson:

- You’ll know the seven pillars cold.
- Each pillar has a one-sentence soundbite *and* a 30-second elaboration.
- You’ll have a full 90-second answer that combines the strongest 3–4 pillars for any given audience.

This is the lesson to re-read the night before any interview. **It is the spine of the whole course.**

1. The frame that holds the seven pillars

Before the pillars, set the frame. Tell the interviewer what you’re about to do:

“AI is genuinely good at producing text that looks like code. Producing text isn’t the same as engineering. There are seven things engineering requires that AI does not do – and probably won’t, in the time horizons that matter for this hiring decision. Let me run through them.”

That intro:

- Concedes AI’s real capability (“good at producing text”) so you don’t sound defensive.

- Distinguishes producing text from engineering (“isn’t the same as”).
- Says “seven” — a specific number. Specificity sells.
- Bounds the claim to “time horizons that matter for this hiring decision” — you’re not making a 100-year prediction, just a 5-year one. **Important.** Don’t let the interviewer drag you into futurism.

Now the seven pillars.

2. Pillar 1 — Problem framing

Soundbite

“AI optimizes the spec it’s given. Deciding what the spec should be is engineering.”

Elaboration (30 seconds)

The hardest part of building software is not writing it. It’s deciding *what* to write. AI tools are excellent at the second half: take a clear specification, produce code. But upstream of that there’s a cloud of ambiguity — *who is the user, what problem are they actually facing, what trade-off should this feature embody, what’s the smallest version that proves the idea.* Those questions don’t have right answers; they have *contextual* answers. AI can suggest options. It can’t pick the option that’s right for **this team, this customer, this constraint.** That’s a human job and it requires sustained engagement with non-engineers.

Concrete example

“On my last project, the PM asked for a ‘scoring system.’ The naive AI prompt would have produced a perfectly working scorer. But the actual job was to choose between three definitions of ‘score’ — points-per-pick, accuracy-percentage, or playoff-style — each of which served a different storyline for the user. I went and watched the league members react to mockups before any code got written. AI didn’t sit in those meetings.”

3. Pillar 2 – Context the AI does not have

Soundbite

*“AI knows code. Engineers know **this** codebase, **this** team, and **this** customer.”*

Elaboration (30 seconds)

AI has the public internet’s worth of code in its head. What it does not have: your codebase’s local conventions, your team’s deploy story, the war story from last quarter when the queue overflowed, the senior engineer who refuses to touch the legacy module, the customer who pays you 40% of revenue and hates change. Those facts shape every real engineering decision. **AI’s suggestions are correct in the average codebase. Your team is not the average codebase.** Adapting AI suggestions to local context — knowing what to keep, what to throw out, what to reshape — that’s the work.

Concrete example

“AI suggested I use Redux for the state management on a new module. In our codebase Redux had been deprecated for 18 months in favor of Zustand. The AI couldn’t know that. A new hire who blindly accepted that suggestion would have wasted a sprint adding a deprecated dependency.”

4. Pillar 3 – Trade-off ownership

Soundbite

“AI can list trade-offs. AI doesn’t make trade-offs, because it doesn’t own consequences.”

Elaboration (30 seconds)

When you make a trade-off — pick A over B — you accept that A’s downsides will surface in production, in code review, in the on-call channel at 2am. The AI does not show up to those moments. It doesn’t get paged. It doesn’t get reviewed. It doesn’t have a reputation that suffers when the trade-off ages badly. Engineering decisions live with their consequences for years. **The accountability for those consequences is the most valuable thing humans**

bring. When the postmortem asks “why did we choose this?”, the AI cannot answer. You can.

Concrete example

“We had to choose between caching at the CDN, the application server, or the client. AI laid out the three trade-offs cleanly. The decision required knowing that our CDN bill was already over budget, that our app servers had spare CPU but tight memory, and that 30% of our users had unstable connections. None of that was in the AI’s prompt. We picked the application-server caching. When it later created a memory pressure issue, I owned the decision — and the fix.”

5. Pillar 4 — Long-horizon judgment

Soundbite

“AI is great at the next 200 lines. Engineering is great at the next 2 years.”

Elaboration (30 seconds)

LLMs operate on a context window. Your project operates on a multi-year arc. Decisions you make now — schema design, API contract, module boundaries — get harder to reverse with every passing month. The skill of saying “this looks fine for the demo but it’ll be technical debt for the next five years, let’s spend an extra hour and do it right” is *temporal judgment*. AI doesn’t have it. It doesn’t even know the current date with confidence. **It cannot weigh present convenience against future maintenance.** Senior engineers do that every hour.

Concrete example

“AI suggested we store user picks as comma-separated strings to keep the API simple. That works at 8 users. At 80 users we’d have been parsing strings on every read. I made the call to use a normalized schema. AI optimized for the prompt; I optimized for two years from now.”

6. Pillar 5 — Cross-team and human translation

Soundbite

“Software is mostly built in conversations. AI is not in those conversations.”

Elaboration (30 seconds)

Engineering work flows through Slack threads with PMs, design reviews with UX, escalations with security, contracts with infra, business reviews with finance. Each of those conversations involves translating between domains: explaining a technical constraint to a PM, explaining a regulatory constraint to an engineer, explaining a deployment risk to a CFO. **Software is the artifact. The conversations are the work.** AI doesn’t participate in those conversations. It can summarize a thread; it cannot persuade a stakeholder, calibrate a roadmap, or notice that the design partner’s actual ask isn’t what they typed.

Concrete example

“PM asked for a feature ‘by Friday.’ I noticed the real constraint was a customer demo on Tuesday. The right move was a smaller scope shipped Monday rather than the full feature shipped Friday. AI doesn’t catch that — there’s no prompt that surfaces a demo two states away.”

7. Pillar 6 — Quality judgment under uncertainty

Soundbite

“AI can write code. Engineers know whether the code is good for our situation.”

Elaboration (30 seconds)

AI output is plausibly-correct on the surface and frequently subtly wrong. Spotting the wrong subset requires **taste** — knowing which patterns wear well, which abstractions leak, which “clean” architectures will fight you in 3 months. Taste isn’t innate; it’s earned by reading a lot of code, shipping a lot of code, and watching it age. AI does not have taste. It has averages. The skill of “this is correct but it’s the wrong correct for us” is irreducibly human.

Concrete example

"AI generated a function that walked the data twice — once to filter, once to map. Functionally fine. For 8 teams. We had a planned migration to 800 in Q2. I rewrote it as a single pass, not because the original was 'wrong', but because I could see the next refactor."

8. Pillar 7 — Accountability and trust

Soundbite

"At the end of every PR is a human signature. The signature is the value."

Elaboration (30 seconds)

When code ships, someone owns it. The person on the call. The person who explains it to the auditor. The person whose name is on the postmortem. AI cannot be that person. Not legally, not socially, not professionally. As long as software has consequences — and it always will, especially in regulated, security-critical, or revenue-bearing contexts — there will be a human at the bottom of the trust chain. **Engineers are the trust layer.** AI tools are below us in the stack, like compilers and linters. Useful, important, not interchangeable with the engineer.

Concrete example

"I shipped a payments change last quarter. The bug it introduced cost us four hours of partial-charge confusion. The remediation involved a written explanation to compliance, a refund script, and a personal apology to a customer. There is no version of that incident response where 'the AI did it' is an acceptable sentence. I owned it because I had to."

9. The 90-second answer (master version)

When you have 90 seconds, here is the answer:

"Sure. AI is genuinely good at producing text that looks like code, and that's a real productivity gain. But producing text isn't the same as engineering. There are seven things engineering requires that AI doesn't credibly do in the next five years.

*First, **problem framing** — deciding what the spec should be in the first place. AI optimizes the spec; humans choose it.*

*Second, **context** — AI knows code, but engineers know this codebase, this team, this customer. The right answer in your shop is rarely the average answer in the AI's training data.*

*Third, **trade-off ownership**. AI can list options. It doesn't accept consequences. The accountability for the choice is what we're hired for.*

*Fourth, **long-horizon judgment**. AI is great at the next 200 lines. We're paid for the next two years.*

*Fifth, **cross-team translation**. Most engineering happens in Slack threads, design reviews, and escalations. AI isn't in those conversations.*

*Sixth, **quality judgment**. AI produces plausibly-correct output. Knowing whether it's the right correct for our context — that's taste, and taste is earned.*

*Seventh, **accountability**. At the end of every PR is a signature. AI can't sign. We can.*

AI is the most powerful tool we've ever had. It raised the floor of what's possible. It hasn't lowered the ceiling of what's required from a senior engineer."

That's about 250 words, ~95 seconds at a normal pace. **Drill it.** It works in any framing.

10. The 30-second answer (compressed version)

When the interviewer is rushing or just wants a soundbite:

"AI is good at producing text. Engineering is more than text. The work is problem framing, trade-off ownership, long-horizon judgment, cross-team translation, taste, and accountability. AI hasn't replaced any of those — it's just made

them more visible by automating the typing underneath. The engineer is what's left when the typing is free."

That's 70 words. Use it when there's no time for the full pillar walkthrough.

11. The 10-second one-liner

For when an interviewer interrupts with the question while you're answering something else:

"AI raised the floor of what's possible. It hasn't lowered the ceiling of what's required."

That's 14 words. Useful as a closer, a hallway answer, a Slack reply to your manager.

12. Customizing the pillars by audience

Not every audience needs all seven pillars. Pick three.

Audience	Lead with	Include	Skip
Hostile	7 (accountability)	3, 4	1, 5
Curious	1 (framing)	5, 6	2
Technical	6 (quality)	2, 4	5
Business	7 (accountability)	3, 5	6
Existential	1 (framing)	5, 7	2, 4

Match the table by remembering the framing from Lesson 1, then picking the three pillars from this row.

13. The pre-emptive objections

Smart interviewers will push back. Have prepared rebuttals.

Objection: “But AI is getting better fast.”

“It is. So is the bar for senior engineers. Five years ago a senior was someone who could ship features fast. Today they have to do that and judge AI output, and design for AI-assisted teams, and evaluate when AI shouldn’t be in the loop. The job got harder, not easier.”

Objection: “Couldn’t I just use a junior plus AI for the price of you?”

“You can. You’ll get a lot of typing-rate productivity. The first time the AI’s suggestion conflicts with a deploy story, a customer constraint, or a security review, the junior won’t have the judgment to push back. That’s the sound of technical debt accumulating quietly. I’d argue you save by hiring me; the bill comes due either way.”

Objection: “What if I told you AI will fully replace developers in 10 years?”

“I can’t disprove a 10-year prediction. I can tell you that none of the AI-replacement timelines I’ve seen come from people who are currently shipping production code with AI tools daily. The people closest to the work are the most measured about its limits. I trust that signal more than I trust the timelines.”

Objection: “Aren’t you just defending your job?”

“Sure, partly. I’d be lying if I said I had no skin in the game. But I’m not asking you to take my word — I’m describing observable, daily failures of AI output that I and every other engineer I know spent today fixing. The case rests on the data, not on my motive.”

That last one is a *judo* move — acknowledging the bias and then pointing to evidence. Highly effective.

14. The body language

Three rules during the seven-pillar answer:

1. **Slow down.** This answer is your highest-value 90 seconds. Don’t rush.

2. **Use your hands.** Count the pillars on your fingers. Visual anchors help interviewers remember.
3. **Pause after each pillar.** Half a beat. Lets the words land.

If you record yourself delivering this answer once, you'll cringe. Do it anyway. Record again. By the third take, you'll be in interview-grade.

15. CHECK YOURSELF

- ☐ Can you list the seven pillars in order, from memory?
- ☐ Can you give the soundbite for each one?
- ☐ Can you give a concrete example for at least four of them — drawn from *your own* work, not the snooker app?
- ☐ Can you deliver the 90-second master answer in under 100 seconds, with no notes, twice in a row?
- ☐ Can you handle the four pre-emptive objections?

If yes to all five, you have the centerpiece of your interview prep locked.

16. Where you are now

You hold the framework that, more than any other single artifact in this course, will get you the offer. Move on to [03-real-world-cases.md](#) — eight short stories that turn the abstract pillars into memorable specifics.